



昆山杜克大学创新孵化器
DKU Innovation Incubator

Dii Final Report

Project Code: IE169

Project Name: Radikal

Team Name: Wision

Team Logo:



Project Summary (50-150 words):

"Radikal" is an entrepreneurship training project developing a gamified, AI-powered mobile application to address the significant challenge of memorizing Mandarin Chinese characters. Unlike rigid, curriculum-based platforms, it allows learners to import vocabulary from any source and utilizes interactive character construction, personalized memory associations, and AI-generated stories to create meaningful, emotional connections with the language.

Targeting tech-savvy young learners globally, the app employs a freemium model, monetizing through AI token purchases and institutional licenses. By transforming rote memorization into an engaging, creative activity, the project aims to make Chinese more accessible, support quality education (SDG 4), and foster greater cross-cultural understanding.

Team Member(s):

Name	Net ID	Class	Major	Role
Bilguun Dugarsuren	bd194	2028	Applied Mathematics and Computational Science in the Computer Science track	Survey and logic flow designer, Developer, Team Treasurer
Nemuulen Togtbaatar	nt152	2027	Computation and Design in Computer Science track	Data collection & survey distributor, Developer, Team Leader, Communications



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Brief Description of Your Project Journey:

1. Entrepreneurial Background, Significance, and Objectives

Radikal was inspired by a common problem we observed among ourselves and our fellow Chinese learners: many students find Chinese characters difficult to memorize because they look visually complex and are often learned through an exhaustive repetition process rather than understanding. In DKU's international environment, we noticed that many students learning Chinese struggled to remember characters, understand radicals, and connect character structure with meaning.

To test whether this problem was meaningful, we asked DKU students who were learning Chinese about their learning difficulties and their willingness to use a more interactive learning tool. Their responses showed that students were interested in a product that could make Hanzi learning easier, more visual, and less repetitive. This feedback gave us confidence that Radikal had real educational value. You can [click here](#) to access the feedback we have collected.

Our objective became to design an AI-enhanced and gamified Chinese character learning platform for young learners. Radikal aims to help users learn characters through radical breakdowns, mnemonic stories, games, points, streaks, tokens, and achievements. As the idea developed, we also saw its business potential through premium AI-supported learning content and a freemium model.

Radikal's long-term vision is a world where Chinese character learning is not a barrier but a driving force. By turning memorization into creation, the app bridges cognitive science (personal story formation strengthens memory) with the cultural richness of the Chinese writing system. Beyond Mandarin, the Vision team plans to adapt the radical decomposition model to Japanese Kanji as the next major product expansion, applying the same AI mnemonic engine to a second major logographic writing system.

2. Market Analysis and Competitive Advantages

Radikal targets young Chinese learners, especially students aged 13–25 who are familiar with mobile apps and gamified learning. Existing tools such as Duolingo, HelloChinese, Pleco, Anki, and Skritter support Chinese learning, but many focus mainly on vocabulary, dictionary search, flashcards, or handwriting practice.

Radikal's advantage is that it combines radical-based learning, AI-generated mnemonic stories, and gamified practice in one mobile-first platform. Instead of treating characters as random shapes, Radikal helps users understand how characters are built and remember them through its building blocks and meaningful mnemonic stories. Its scalable database and flexible product design also allow future expansion into audio support, classroom tools, and premium learning services.



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Table 1. SWOT analysis

Strengths	Weaknesses
-Engaging Gamified Platform: Our app leverages interactive and competitive game mechanics that align with the preferences of younger users, driving engagement and improving retention rates. -AI-Powered Storytelling: By adapting content to individual learning styles and progress, the app offers a customized experience that enhances language acquisition efficiency. -Strong Market Fit: Targeting a tech-savvy demographic with high digital engagement provides a solid foundation for rapid adoption. -Highly flexible and Personalized: Making it compatible for classroom learning and individual learning allows more market potential.	-Limited Brand Awareness: As a new entrant, the app may face challenges in establishing visibility and trust within the competitive language-learning market. -Resource Constraints: Developing and maintaining advanced gamification features and personalized content may require significant investment in technical resources and content creation.
Opportunities	Threats
-Growing Demand for Mandarin Learning: With an estimated 15 million young learners actively pursuing Mandarin, there is a substantial and expanding market to capture. -Social Learning Integration: Incorporating collaborative and competitive features can drive viral growth and increase user engagement. -Partnerships with Schools and Universities: Collaborations with educational institutions can accelerate adoption and expand the app's reach.	-Intense Competition: Established language-learning platforms like Duolingo, Memrise, and Chineasy pose strong competition with larger user bases and brand recognition. -Technological Shifts: Rapid changes in digital learning trends may require continuous updates and feature enhancements to stay relevant.

Business Plan

1. Product design

Core Concept

Radikal is built on one insight: learners remember characters they have a story for. The app decomposes every character into its radicals, and words into characters, then uses AI to generate a short, punchy mnemonic story that links radical meanings to the character's meaning and sound. Users can regenerate, edit, or write their own story.

Learning Modes

Radikal has six learning modes, each targeting a distinct cognitive skill:



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- Guess the Hanzi: flashcard-style quiz from mnemonic or definition to character
- HanziLab: assemble radicals to construct real characters
- Class Mode: teacher-managed decks and student progress tracking
- Focus Mode: spaced-repetition for characters due for review
- Dictionary: paginated browse and full-text search of 4,953 characters and 9,424 words
- Community: leaderboard, XP milestones, and community forum

AI Mnemonic Engine

Mnemonic generation is powered by the Kimi/Moonshot API (model: moonshot-v1-8k) via a Supabase Edge Function. The engine applies strict quality principles:

- 1-3 sentences maximum; punchy over literary
- All radicals are cited with their English meaning in parentheses
- No lazy pinyin introductions; sound is embedded naturally
- Single-radical characters describe visual shape, not composition

Technical Architecture

The current production stack:

- Frontend: React 18 + Vite 6, Tailwind CSS, Shadcn/Radix UI
- Backend: Supabase (PostgreSQL, Edge Functions, Auth, Storage, Realtime) on Singapore region
- Mobile: Capacitor (iOS + Android), distributed via TestFlight; App Store release pending
- Hosting: Netlify (SPA, Node 20)
- Payments: LemonSqueezy (frontend wired; backend implementation upcoming)
- Character database: 18,958 rows (4,953 characters, 9,424 words, 4,581 unclassified); 218 radicals; sourced from Unihan, CC-CEDICT, Make Me a Hanzi, ivankra/hsk30

2. Business model

Model Overview

Radikal operates as a Software-as-a-Service (SaaS) product with a freemium revenue model. The free tier provides sufficient value to grow the user base organically. Monetization occurs through AI token purchases, institutional subscriptions, and in-app advertising.

Revenue Streams

- AI Token Sales: each mnemonic generation consumes one token; tokens sold in packs or earned through ads/streaks
- Institutional Licensing: time-based, unlimited access sold to universities and language schools per student
- In-App Advertising: free-tier users earn tokens by watching ads; standard display advertising revenue



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Pricing Structure

Table 2. Pricing Structure

Tier	Features	Price	Target
Free	Core learning modes, 3 tokens/day, ad-supported	\$0	Individual learners
Starter Pack	50 tokens for mnemonic generation	6.99 CNY/ \$0.99	Independent learners
Explorer Pack	100 tokens for mnemonic generation	9.99 CNY/ \$1.49	Independent learners
Master Pack	200 tokens for mnemonic generation	19.99 CNY/\$2.99	Independent learners
Institutional	Unlimited access per student, 6-month term	\$6 / student	Schools & universities

Streak bonuses award 3 free tokens daily to active users. Ad-watching awards 3 tokens per view. This keeps the free experience meaningful while incentivizing upgrades.

App Distribution

- iOS: Xcode / Capacitor; currently TestFlight; App Store submission in progress
- Android: Google Play Store (Capacitor Android shell)
- Web: Radikal.one (Netlify, PWA-capable)

Compliance

GDPR-compliant Terms and Conditions and Privacy Policy are drafted and in legal review. App Store certification and compliance vetting are to be done soon.

3. Marketing plan

Target Audience

Primary: independent Mandarin learners aged 13-25, globally distributed, tech-native, active on social media.

Secondary: Chinese language teachers and university language programs.



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Go-To-Market Strategy

Phase 1: Institutional Seeding

Partner with DKU and peer universities offering Chinese language courses. Provide institutional access to faculty and students. Use this cohort for product feedback, testimonials, and case studies.

Phase 2: Community-Led Growth

Launch social campaigns leveraging user-generated mnemonic stories. Campaigns include:

- #MyFunMemo: users share creative character mnemonics publicly
- Memorization Challenge: AI generates 10 characters; users film reaction videos showing recall
- Teacher Ambassador Program: compensate language teachers on YouTube, TikTok, and Instagram for tutorials

Phase 3: Paid Acquisition

Performance marketing on Instagram, TikTok, YouTube, and Reddit language learning communities (r/ChineseLanguage). SEO targeting 'learn Chinese characters' and related high-intent queries. App Store Optimization (ASO) for iOS and Android.

Positioning

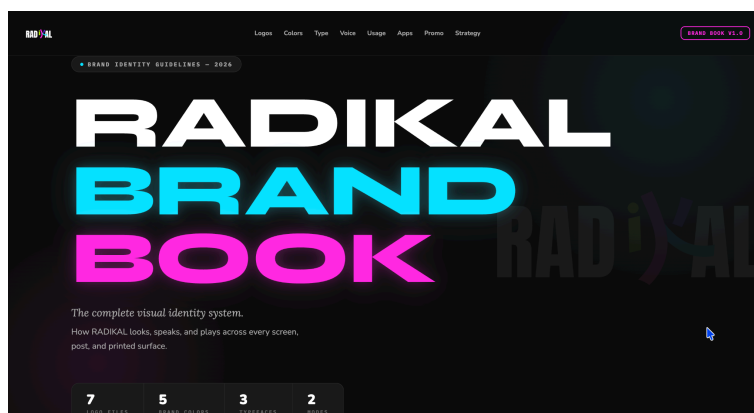
Radikal is positioned as a supplement to existing Chinese learning methods, not a replacement. This avoids direct head-to-head competition with Duolingo and Memrise, and increases integration opportunities with institutional curricula.

Brand Identity

The Radikal brand uses a dark-first aesthetic with Radical Pink (#FF2D78), Glitch Cyan (#00D4FF), and Acid Green (#C8FF00) as signature colors. Typography: Lora Italic for personal moments, Nunito for UI, IBM Plex Mono for technical labels. The logographic wordmark 'Radi火II' anchors the visual identity.

Brandbook link: <https://duke.box.com/s/5080zybcaovgb6519eza79irc9inc9ba>

Logo and graphic assets: <https://duke.box.com/s/hduxslrrf9ql2yps3upvag01j4febdpn>





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4. Financial Forecast & Revenue Projections

The following table outlines the projected revenue growth from Year 1 to Year 4 based on user acquisition and monetization strategies.

Table 3. Revenue Projections

Year	Free Users	Paying Users (5%)	Token Revenue	Inst. Revenue	Ad Revenue	Total Revenue
1	500	0	\$0	\$0	\$100	\$100
2	10,000	500	\$1,000	\$3,000	\$2,000	\$6,000
3	50,000	2,500	\$5,000	\$15,000	\$10,000	\$30,000
4	125,000	6,250	\$12,500	\$37,500	\$25,000	\$75,000

Note: Token revenue is calculated at an average of ~\$2 per paying user per year. Institutional revenue is calculated at \$6 per student.

Table 4. Profit / Loss Projections

Year	Total Revenue	Total Costs	Profit / Loss
1	\$100	\$4,900	-\$4,800
2	\$6,000	\$5,000	+\$1,000
3	\$30,000	\$10,000	+\$20,000
4	\$75,000	\$10,000	+\$65,000

Key Assumptions

- Addressable Market:** Approximately 15 million young Mandarin learners (ages 10–25) globally.
- Market Share Goal:** 2% long-term target, equating to 300,000 active users.
- Conversion Rate:** Estimated 5% of active users will convert to paying tiers.
- Pricing Model:** Token packs priced between ¥6.99 and ¥19.99 (~\$1–\$2.80); Institutional licensing at \$6/student per term.
- Institutional Growth:** Revenue expected to accelerate in Years 2–3 via DKU and partner university partnerships.



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6. **Profitability:** Break-even achieved between Years 2 and 3 through infrastructure automation and low overhead.

Project Implementation

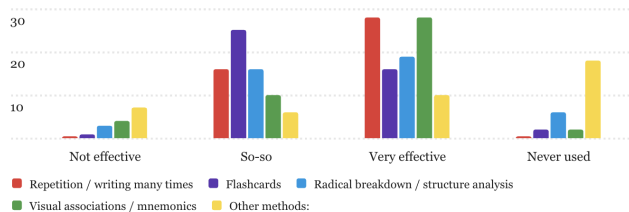
Market Research

We began by observing the problem firsthand among our peers at DKU before conducting any formal research. After building an early working prototype, we distributed a structured survey to DKU students currently learning Mandarin. 63 students consented and 44 completed the full survey. Results confirmed that 48% of respondents identified character memorization as a top 2 difficulty, and open-ended responses repeatedly requested radical breakdowns, mnemonic stories, and interactive practice, thus validating both the problem and our proposed approach.

Primary market research:

Figure 1. Primary market research: Effectiveness of memorization methods

Q8 - Please rate the effectiveness of each of these memorization methods.



Insights from the short quiz:

Q14 - What was the definition of the character "囧"



Q15 - What was the definition of the character "呆"



Q16 - What was the definition of the character "鲁"



SC0 - Score

Field	Min	Max	Mean	Median	Responses
Score	1.00	3.00	2.82	3.00	38

Insight: The mean score is 2.82, confirming our method is very effective.

During our early-prototype demonstration events on campus, we collected live feedback on usability, learning-mode engagement, and mnemonic quality, which directly informed later improvements to the game mechanics and overall user experience. Please go to the following link to see the full research reports and analysis:

<https://duke.box.com/s/ll0i5mmyf9vd8bz1fu3r36pqakh1ba99>

Secondary market research and competitor analysis:



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Figure 2, 3. Direct and Indirect Competitor Analysis

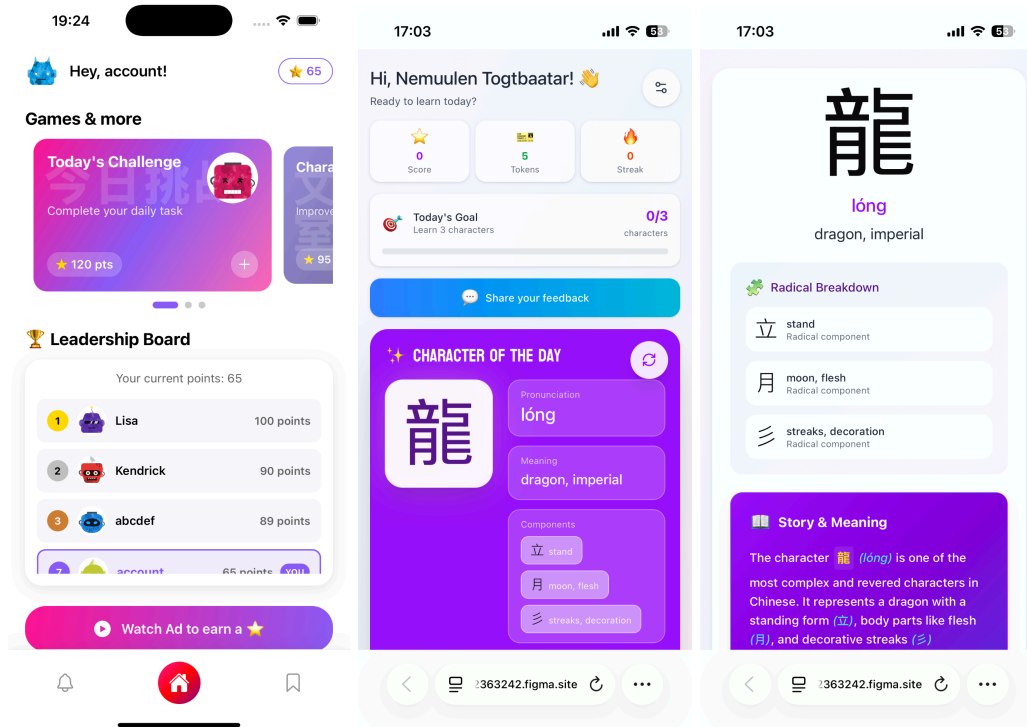


Product Development

The first working prototype was built on Base44 and Figma to validate core interactions quickly, and was used at our first on-campus demonstration event to gather feedback before committing to a full engineering build.

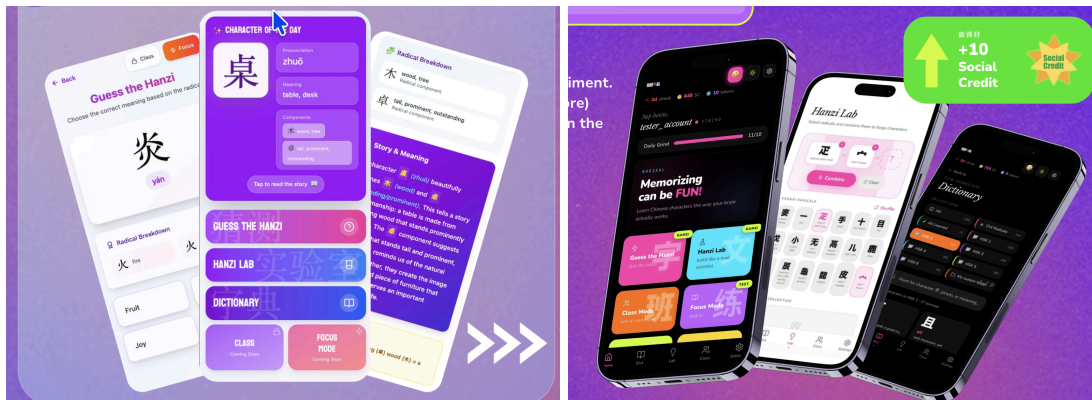
We then migrated to AWS for a full-stack rebuild. However, running AWS proved costly, and the architecture was more complex than our stage required. In parallel, we had already been prototyping the AI mnemonic generation function directly on Supabase Edge Functions. Migrating to Supabase was therefore both the simpler and more cost-effective path. We moved the entire backend to Supabase in the Singapore region, which also resolved the latency issues users had reported during AWS-hosted beta testing, and became the stable production foundation the app runs on today.

Figure 4. Application interface examples





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Product Testing, Feedback, and Ongoing Development

The first demo event used the early Base44 prototype. Student feedback identified two clear issues: mnemonic stories were too long, and the home page needed to be more intuitive. This directly produced the quality rules now built into the AI engine and the UI.

The second event showcased the full production application, and feedback was significantly more positive, with several students creating accounts during the session. Responses informed specific improvements to the HanziLab forge matching logic and hint flow.

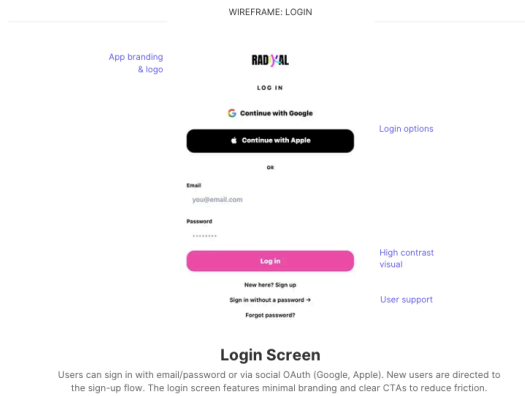
The XJTLU language fair extended our reach beyond DKU for the first time and produced the most concentrated round of feedback. During the expo itself, we encountered several bugs that had been overlooked during debugging. Following the event, we worked through these issues systematically. Faculty at XJTLU also raised feature suggestions around Class Mode and institutional use. These feedback sessions were invaluable and continue to inform the product roadmap as we approach the public App Store launch. As we build connections with the XJTLU faculty, we plan to continue collaborating with them and present our institutional plan to them.

Final Product

Figure 5. Application wireframes and onboarding flow

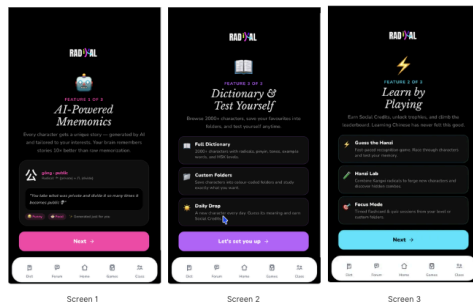


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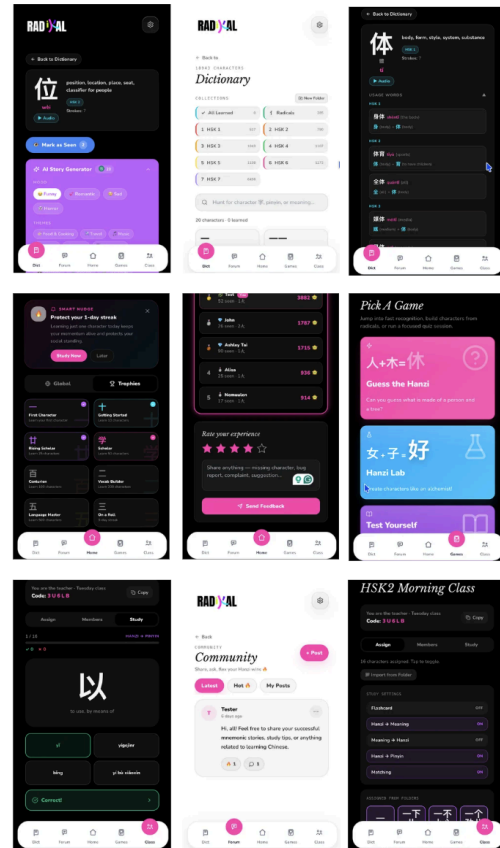
Login Screen

Users can sign in with email/password or via social OAuth (Google, Apple). New users are directed to the sign-up flow. The login screen features minimal branding and clear CTAs to reduce friction.



Onboarding Flow

A 3-step onboarding carousel introduces new users to core features. Each screen highlights one value proposition with a simple illustration, headline, and short description. Users can skip at any time or tap through sequentially. The final screen's CTA transitions to account creation.



Currently, Radikal offers six learning modes, each designed to strengthen a different cognitive skill. Guess the Hanzi uses flashcard-style quizzes that help learners recall characters from mnemonics or definitions. HanziLab encourages structural understanding by allowing users to assemble radicals and construct real characters. Class Mode supports teacher-managed decks and student progress tracking, while Focus Mode uses spaced repetition to help learners review characters at the right time. The Dictionary provides paginated browsing and full-text search across 4,953 characters and 9,424 words. Finally, the Community mode motivates continued engagement through leaderboards, XP milestones, and a discussion forum.

Team

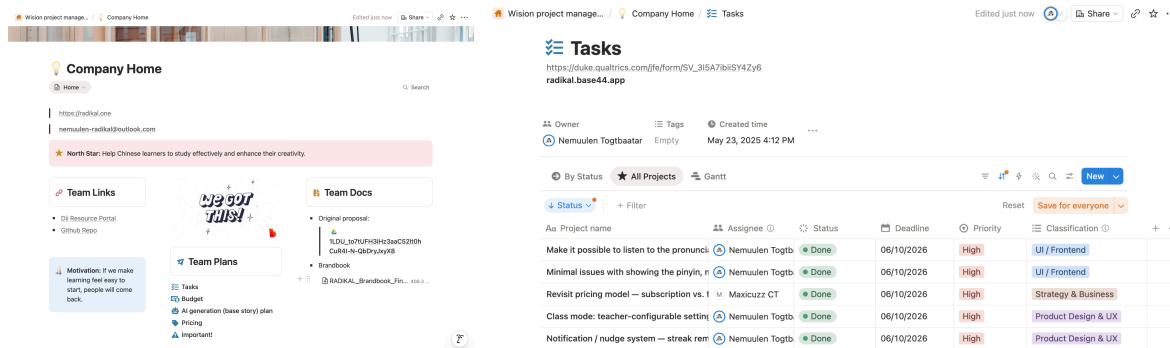
Responsibilities are divided along complementary strengths: product design, user research, communications, and event organisation on one side; system architecture, logic flow, backend development, and financial tracking on the other. Both founders contributed to frontend development, AI prompting, and database construction throughout. The main operational constraint was the absence of a dedicated person for marketing promotional activities, and development competed for the same limited time, which will be addressed in the next phase.



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We used Notion for project planning, task allocation, timeline management, and documentation. It helped us organize weekly goals, assign responsibilities among team members, track progress, and keep all research notes, design decisions, and development updates in one shared workspace.

Figure 6. Project management dashboard



Outcomes and Major Achievements (Evidence-Based)

- Launched a fully functional mobile web application ([Radikal.one](https://radikal.one)) built on a production Supabase backend with an 18,958-entry character and word database, six learning modes, AI-generated mnemonic stories, XP/streak gamification, and a community feed – deployable to iOS and Android via Capacitor.
- Conducted 2 on-campus early- prototype demonstration events with over 30 student participants each, collecting live feedback on usability, learning mode engagement, and mnemonic quality; feedback was used to directly iterate on the game mechanics and user experience.

Figure 7. Prototype demonstration event



[Click here](#) to see the materials we have used for these events.

- Ran 2 promotional tabling sessions on campus, reaching prospective users with live app demos and gathering sign-up interest from Chinese language learners across year groups.

Figure 8. Promotional tabling events



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- Traveled to an external university (XJTLU) Chinese language fair to present Radikal, receiving positive reception and comments from attendees and faculty; the event validated cross-institutional demand for a digital memorization supplement.

Figure 9. External university language fair presentation



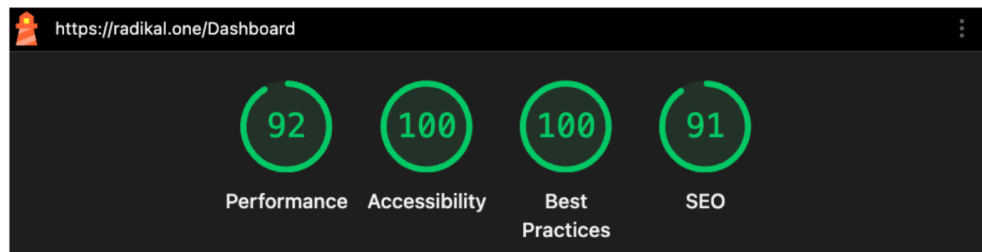
- Received consistently positive qualitative feedback across all events, with attendees highlighting the mnemonic story feature and gamified character-building as standout differentiators from existing tools like Duolingo.
- Implemented end-to-end payment infrastructure with LemonSqueezy, enabling token purchases and institutional subscription flows directly within the app.
- Shipped a fully working AI mnemonic generation feature with high user productivity, where learners can generate, regenerate, and personalize memory stories for any character in seconds, powered by a live API integration with quality-controlled prompting.
- Website performance and accessibility are rated good according to the Chrome Lighthouse tool.



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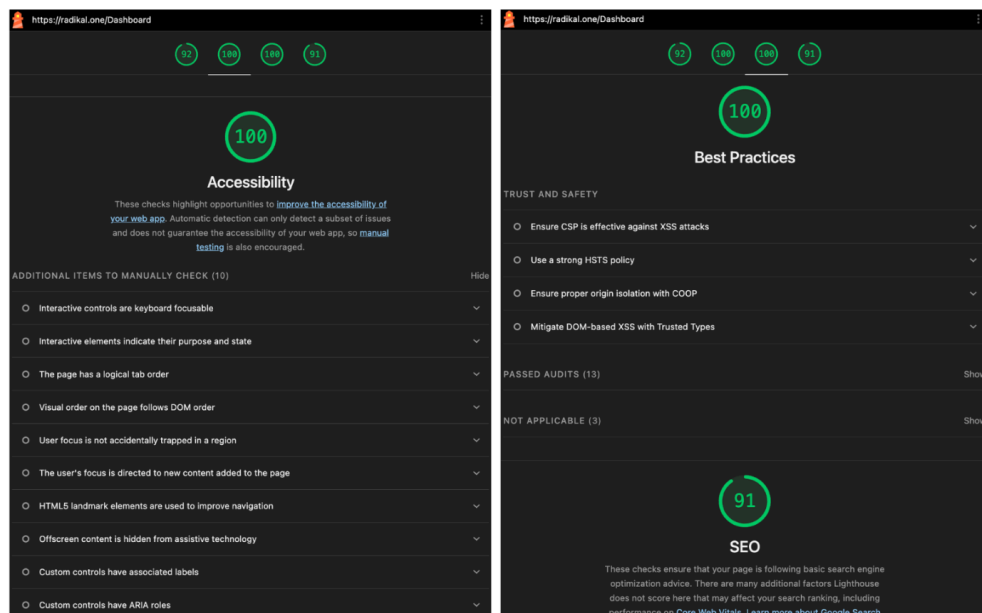
Figure 10. Website performance analysis

WEBSITE HEALTH OVERVIEW: DASHBOARD



Website Health Overview

Our site demonstrates a good level of technical performance and optimization. With Performance at 92, Accessibility at 100, Best Practices at 100, and SEO at 91, the dashboard confirms that the platform is fast, secure, user-friendly, and highly visible to search engines. These scores reflect strong compliance with modern web standards, ensuring a seamless experience for users while maintaining long-term sustainability.



Why These Scores Matter

Performance: A fast-loading site improves user experience and reduces bounce rates.

Accessibility: Ensures inclusivity, allowing all users, including those with disabilities, to navigate and interact seamlessly.

Best Practices: Strong security and coding standards protect users and maintain trust.

SEO: Optimized visibility helps the site rank higher in search engines, driving more organic traffic.

- We also created an introductory website for Radikal, aimed at institutions, educators, researchers, and potential collaborators. The website presents Radikal through real app screenshots, brand-based visuals, and smooth product animations, helping position the project



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as a polished, mobile-first learning platform ready for classroom pilots, research partnerships, and educational collaboration.

See here: <https://silly-portal-14927370.figma.site>

Box link containing all the evidence above:

<https://duke.box.com/s/6wxnn2sje8qdohjtk8xx945fpq3izzr>

Note:

All outcomes must be supported by **concrete evidence**, such as:

- Quantitative data (e.g., performance metrics, user numbers, improvement rates)
- Qualitative insights (e.g., interviews, testimonials, observation records)
- Tangible deliverables (e.g., prototypes, product iterations, posters, reports). Acceptable forms of evidence include demo links, GitHub repository links, and screenshots or video recordings submitted via the designated Box folder.
- Process documentation (e.g., experiment logs, iteration records, workshop materials)

Describe the innovative features, practical significance, and social impact of the project.

Gains and Experiences:

Looking back, what stands out most is how Radikal grew from a problem we personally felt into something we genuinely believe in. Its core innovation is turning Chinese character learning from rote memorization into a visual, interactive, and meaning-based experience, and the strength we are proudest of is how closely it stays tied to a real learning need. Our early prototype demonstrations and student feedback confirmed what we suspected from our own studies: many learners struggle to memorize characters and want a more visual, gamified tool. Over time, the product also grew beyond a dictionary or flashcard app into a fuller learning system, including radicals, mnemonics, spaced review, teacher-managed study, and community, with AI-generated stories as the feature we feel gives it real educational value.

Technically, the project taught us an enormous amount. Neither of us had built something at this scale, and we learned how the layers of a real application actually fit together, how the frontend and backend communicate, how the AI mnemonic generation is wired into the backend, how web hosting works, and how a website can be migrated into a functioning mobile app. Seeing these pieces interconnect, rather than reading about them in isolation, was one of the most valuable parts of the experience. We also learned about the parts of building a SaaS business we had overlooked at the idea stage, most clearly, user privacy. Once we realized the app would collect personal information, we understood that proper terms, conditions, and a privacy policy were a responsibility, not an afterthought.

Working as a two-person team was its own lesson. The two of us have different schedules and lifestyles, so keeping the project moving forced us to communicate efficiently and divide responsibilities along our strengths, for example, one of us handling finance and backend decisions while the other took on privacy, policy, and the frontend, then keeping all of it coherent as one product. These habits of clear



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communication, role division, and follow-through are exactly the kind of experience we expect to rely on in future ventures.

We also faced real challenges. Early AI-generated mnemonics were often too abstract or hard to connect with the character, which we addressed by refining the prompts, simplifying the story format, and acting on user feedback. Additionally, choosing the optimal LLM model was an important part of the improvement. Managing a large character and word database while keeping the app fast led us to improve the dictionary with pagination and search, authentication, database integration, and payment setup, each of which took repeated testing to get right.

Some weaknesses remain. Radikal still needs testing with a larger, more diverse group to measure whether it truly improves long-term retention, the AI content needs stronger quality control, and features like Class Mode, Community, and progress analytics could be developed further. Going forward, we want to run a formal learning-effectiveness study against traditional methods, improve personalization by recommending characters based on each learner's level and mistakes, and keep strengthening the classroom and community functions. Overall, we believe Radikal has strong potential as an educational technology project, though it still needs further testing, refinement, and scaling to become a fully mature product. We have also been awarded a research grant of RMB 20,000 from DKU's Center for Teaching and Learning (CTL), which would support further user testing, learning-effectiveness evaluation, and continued development.

Financial Statement

Within the duration of the program, our spending fell into three categories aligned with the project's needs: (1) AI and development tools, (2) hosting, database, and platform services, and (3) promotion, research, and user feedback. The tables below itemize every expense incurred from February to June 2026. All amounts are in Chinese Yuan (¥).

Date	Item	Category	Purpose	Amount (¥)
Feb	Base44 Subscription (v3)	AI & Dev Tools	Rapid prototyping and early app development	344.20
Feb	Claude Pro	AI & Dev Tools	Programming and research assistance	137.68
Mar	Base44 Subscription (v3)	AI & Dev Tools	Continued prototyping and app development	344.20
Mar	Claude Pro	AI & Dev Tools	Programming and research assistance	137.68
Apr	Apple Developer Program	Hosting & Platform	Subscription required to distribute the app on the App Store	693.00



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Apr	AWS	Hosting & Platform	Backend cloud hosting services	583.00
Apr	ChatGPT Plus	AI & Dev Tools	Development and research	140.00
Apr	Claude Pro	AI & Dev Tools	Programming and backend development	140.00
Apr	Coco Gift Cards (10 × ¥10)	Promotion & Research	Incentives for beta testers to test and give feedback	100.00
Apr	Netlify	Hosting & Platform	Frontend hosting	63.00
May	Claude Pro	AI & Dev Tools	Programming and app building	142.80
May	ChatGPT Plus	AI & Dev Tools	Research and app development	142.80
May	Netlify	Hosting & Platform	Frontend hosting	64.26
Jun	Kimi (Moonshot) API	AI & Dev Tools	AI mnemonic generation tokens (production engine)	1,500.00
Jun	Google Developer Registration	Hosting & Platform	Access to publish on the Google Play Store	175.00
Jun	Claude Pro	AI & Dev Tools	Coding, development, and analysis	140.00
Jun	Claude API credits	AI & Dev Tools	Pre-generated stories and high-load generative tasks	140.00
Jun	ChatGPT Pro	AI & Dev Tools	Project refinement, analysis, and programming	140.00
Jun	AWS	Hosting & Platform	Backend testing	25.48
Total				5,153.10

Category	Amount (¥)	Share
AI & Development Tools	3,449.36	66.9%
Hosting, Database & Platform Services	1,603.74	31.1%
Promotion, Research & User Feedback	100.00	1.9%
Total	5,153.10	100.0%

1. AI and Development Tools - ¥3,449.36 (66.9%)

The largest share of funding supported the technical development of RADIKAL. Because the project required both frontend and backend work, we relied on generative-AI subscriptions and



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coding-assistance tools (Claude Pro, ChatGPT Plus/Pro) for rapid prototyping, debugging, and accelerated technical learning, and on the Base44 platform to build our first working prototype. The single largest line item was the Kimi (Moonshot) API (**¥1,500**), which powers the production AI mnemonic engine, supplemented by Claude API credits for high-load generative tasks. These tools did not replace our own work; they helped us work more efficiently and turn our concept into a functioning product faster.

2. Hosting, Database, and Platform Services - ¥1,603.74 (31.1%)

The second category covered the infrastructure needed to build, host, distribute, and test the product. This included backend and cloud services (AWS), frontend deployment (Netlify), and the developer-program fees required to publish the app, the Apple Developer Program (**¥693**), and Google Developer registration (**¥175**). These services supported authentication, database management, AI integration, and public access to the prototype. This spending was essential, since RADIKAL could not be tested as a real digital product without a stable technical environment.

3. Promotion, Research, and User Feedback - ¥100.00 (1.9%)

The remaining funds supported user research and feedback collection, specifically gift-card incentives (¥100) that encouraged beta testers to thoroughly test the app and provide structured feedback. This evidence from real learners was central to validating both the problem and our solution. Promotional and design materials were produced using Canva Pro under a free trial, so they carry no cost in this statement.

Summary

Total grant expenditure to date is **¥5,153.10**. Roughly 98% went to development tools and infrastructure, reflecting RADIKAL's current pre-revenue, build-and-validate stage. As noted earlier, the app has not yet generated stable commercial income; our financial priority remains using the grant responsibly to improve the product, validate user demand, and prepare for future revenue through token packs, subscriptions, and institutional partnerships.

Acknowledgements

We would like to sincerely thank our supervisor for her guidance and support throughout the Radikal project. As a Mandarin instructor for international students, her professional experience helped us understand our product from a teacher's perspective. Her insights gave us a clearer direction for designing learning content, improving delivery methods, and making Radikal more useful for real Chinese learners.

We also deeply appreciate her help in creating opportunities for us to present and test our product. She supported our first demo launch, helped us reach DKU international students in Chinese classes, and connected us with Mandarin teaching professors from other universities. Through these opportunities, we received valuable feedback from both learners and educators, including during a language fair at



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another university in China.

Beyond academic and professional support, her encouragement during meetings gave our team confidence, especially during difficult stages of development. Her guidance, connections, and positive support played an important role in helping us validate, improve, and continue developing Radikal.

We would also like to sincerely thank DKU's Innovation and Entrepreneurship Initiative (INE) and the Dii team for their continuous support throughout the project. Their workshops, meetings, and practical advice helped us better understand how to develop Radikal not only as an educational tool, but also as a real product with social value and future growth potential. Through their guidance, we gained useful knowledge about project planning, product positioning, user research, pitching, and implementation. Their encouragement and support helped us refine our ideas, improve our project direction, and move forward with greater confidence.

Supervisor Comments:

The Wision Team has successfully completed the project goals. They transformed a real-world question into a functional app and demonstrated great commitment throughout this valuable journey. They also grew significantly as developers, learners, and a team while overcoming challenges. I see great potential in this project and am very proud of their achievements. I look forward to continuing our collaboration to further develop the project beyond the funding period.

Supervisor Signature:

Date: June 13th, 2026